Context-Aware Activity Recognition for Elderly Fall Risk Detection

R26-IT-051

Project Proposal Report

Chathuri M T K – IT22548078

BSc. (Hons) Degree in Information Technology Specialize in Information Technology

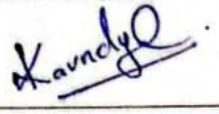
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

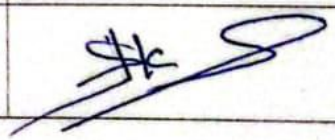
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Chathuri M T K	IT22548078	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Mr.Uditha Dharmakeerthi	06/07/2026	

Abstraction

Falls in the elderly are a global health care challenge, resulting in serious injuries, hospitalizations and increased mortality rates. According to the World Health Organization, falls are the second leading cause of accidental death worldwide, with adults over 65 years of age being the most at risk. Many existing systems address this problem reactively, identifying a fall only after it has occurred and alerting caregivers through simple character-driven mechanisms. These solutions fail to prevent falls and provide limited options for medical personnel.

The identified research gap lies in the lack of proactive prediction-based systems that can anticipate a fall before it occurs and explain the predicted risk in a way that is understandable to humans. Current commercial and research solutions predominantly rely on threshold-based motion detection or post-event classification, lacking real-time predictive intelligence and interpretability.

This research proposes an IoT-based and machine learning-driven system designed to monitor elderly body movements in real time, analyze posture changes and predict imminent fall risk before the event occurs. A wearable IoT device continuously captures body position data; a machine learning model then classifies activity states (walking, sitting or stationary) and predicts the probable fall direction — left, right, front or back. The system further generates a confidence score for each prediction and monitors whether the individual stabilizes, reducing false positive interventions. A key novelty of this research is the integration of Explainable AI (XAI) techniques to produce human-understandable explanations of each prediction, enabling caregivers and clinicians to trust and act on system outputs effectively.

Technologies to be employed include IoT sensors (accelerometers, gyroscopes), edge computing modules, machine learning algorithms (LSTM, Random Forest) and XAI frameworks such as LIME or SHAP. The expected outcome is a wearable fall-prediction system with measurable accuracy and real-world clinical utility.

The system is anticipated to deliver significant social impact by reducing fall-related injuries among the elderly and alleviating the burden on healthcare systems. Commercial opportunities include partnerships with elder care facilities and wearable health-tech providers.

Keywords: Elderly fall prediction, IoT wearable, machine learning, Explainable AI, real-time posture monitoring

Table of Contents

1. INTRODUCTION 1
1.1 Background and Context 1
1.2 Literature Review 1
1.3 Research Gap 3
2. OBJECTIVE 3
2.1 Main Objective 3
2.2 Specific Objectives 3
3. METHODOLOGY 4
3.1 Research Approach 4
3.2 Technical Methods and Algorithms 4
3.3 Tools and Platforms with Justification 5
3.3.1 Frontend Design and User Experience (UX) Strategies 6
3.4 Data Collection and Ethical Considerations 6
3.5 Validation Strategy and Performance Metrics 6
4. HIGH-LEVEL SYSTEM ARCHITECTURE 6
4.1 Overview of Proposed Solution and its Components 6
4.2 System Diagram of Components 6
5. USER REQUIREMENTS 8
5.1 Stakeholder Identification ..... **Error! Bookmark not defined.**
5.2 Functional Requirements (FR) 9
5.3 Non-Functional Requirements (NFR) 10
6. COMMERCIALIZATION PLAN 11
6.1 Target Market and Customer Persona 11
6.2 Value Proposition 11
6.3 Revenue Model 11
6.4 Cost Estimation (Initial Phase) 11
6.5 Pricing Strategy 12
6.6 Competitive Advantage 12
6.7 Intellectual Property (IP) Considerations 12

7. WORK BREAKDOWN STRUCTURE (WBS) 13
8.GANTT CHART 14
9.REFERENCES 15
10.APPENDICES 15

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
IoT	Internet of Things
ML	Machine Learning
LSTM	Long Short-Term Memory
SHAP	SHapley Additive exPlanations
LIME	Local Interpretable Model-agnostic Explanations
WHO	World Health Organization
SDG	Sustainable Development Goal
SVM	Support Vector Machine
RMS	Root Mean Square
API	Application Programming Interface
MCU	Microcontroller Unit
OTA	Over-The-Air (update)
MQTT	Message Queuing Telemetry Transport
TLS	Transport Layer Security
DSR	Design Science Research
PERS	Personal Emergency Response System
SaaS	Software as a Service
HAR	Human Activity Recognition
PCB	Printed Circuit Board

1. INTRODUCTION

1.1 Background and Context

Falls among elderly individuals represent one of the most prevalent and serious health concerns globally. The World Health Organization (WHO) reports that approximately 684,000 fatal falls occur each year worldwide, with adults aged 65 and older accounting for the greatest number of fall-related fatalities and hospitalizations [1]. In Sri Lanka, as the proportion of the elderly population continues to grow, fall-related injuries impose an increasing burden on the national healthcare system and family caregivers.

The consequences of a fall extend beyond physical injury. They include psychological impacts such as fear of falling again, reduced mobility, loss of independence and deterioration in quality of life. Additionally, the economic cost of treating fall-related injuries is substantial, placing strain on both families and public health institutions.

Current solutions in the market -including personal emergency response systems (PERS), wearable accelerometer-based detectors and camera-based surveillance are predominantly reactive. They detect a fall after it has already occurred and trigger an alert. While useful for emergency response, these systems do not address the fundamental challenge: preventing the fall in the first place. Furthermore, existing solutions lack the capacity to provide contextual, interpretable predictions that clinicians and caregivers can act upon proactively.

This research aligns with the United Nations Sustainable Development Goal 3 (SDG 3: Good Health and Well-being), which aims to promote healthy lives and well-being for all ages. By developing a proactive, intelligent fall-risk detection system, this work contributes directly to reducing preventable injuries and healthcare costs associated with elderly falls

1.2 Literature Review

A substantial body of research has explored elderly fall detection and prevention over the past two decades. The following review highlights key approaches, existing products and their limitations.

Threshold-based Systems: Early wearable fall detectors relied on simple threshold-based algorithms applied to accelerometer data. While computationally efficient, these systems are prone to false positives (misclassifying vigorous activity as a fall) and false negatives (missing slow-motion falls) [2].

Machine Learning Approaches: Subsequent research adopted machine learning techniques including Support Vector Machines (SVM), Random Forests and Neural Networks to improve classification accuracy. Studies such as [3] demonstrated that LSTM networks effectively capture temporal patterns in body movement data, achieving above 90% accuracy in fall detection benchmarks. However, these models primarily focus on post-fall classification rather than pre-fall prediction.

Camera-Based Systems: Vision-based systems using depth cameras and pose estimation (e.g., Kinect) offer rich contextual data but raise significant privacy concerns for in-home deployment and require complex installation [4].

Existing Commercial Products: Solutions such as the Apple Watch fall detection, Philips Lifeline and Medical Guardian detect falls post-event and send alerts. None of these provide pre-fall prediction or explainable outputs for clinical use [5].

Explainable AI in Healthcare: The application of XAI methods such as LIME and SHAP in healthcare predictive models has grown significantly, improving trust and clinical adoption [6]. However, their integration into real-time wearable fall prediction systems remains largely unexplored.

Posture Analysis and Directional Prediction: A limited number of studies have explored predicting the direction of an impending fall. Research in [7] demonstrated that gyroscope and accelerometer fusion can characterize postural instability preceding a fall, but directional prediction and confidence scoring were not addressed.

In summary, the literature reveals that while fall detection has matured considerably, proactive fall prediction with directional intelligence, confidence scoring and explainability remains an open research problem.

1.3 Research Gap

The review of existing literature and commercial solutions reveals three critical gaps that this research addresses:

1. Reactive rather than proactive: Virtually all existing systems detect a fall after it has occurred. No widely deployed system predicts an imminent fall before impact, which is the only window during which injury can be prevented.
2. Absence of directional fall prediction: Current systems do not predict the direction in which a fall is likely to occur (left, right, front, back). Such information would be invaluable for targeted intervention and rehabilitation design.
3. Lack of interpretability: Machine learning-based fall prediction models are predominantly black-box systems. The absence of human-understandable explanations limits clinical trust and adoption.

This research addresses all three gaps by developing a proactive, directional and explainable fall-risk prediction system - a combination not previously demonstrated in the literature.

2. OBJECTIVE

2.1 Main Objective

To develop an IoT-based, machine learning-driven system that predicts the risk of falls in elderly individuals in real time - before the fall occurs by analyzing body movement patterns and postural changes, while providing human-understandable explanations of each prediction.

2.2 Specific Objectives

1. To design and implement a wearable IoT device integrating accelerometer and gyroscope sensors capable of continuously capturing body movement and positional data at sufficient sampling frequency.
2. To develop and train a machine learning model (LSTM/Random Forest) that classifies real-time activity states (walking, sitting, standing) and predicts the probability and direction of an imminent fall.
3. To implement a confidence scoring mechanism that quantifies the certainty of each fall-risk prediction and monitors whether the subject stabilizes after an at-risk state.
4. To integrate Explainable AI (XAI) techniques (LIME or SHAP) to generate natural language or visual explanations of fall-risk predictions understandable to caregivers and clinicians.
5. To evaluate the system's performance in terms of prediction accuracy, precision, recall, F1-score and response time using real-world or benchmark datasets.
6. To validate usability and acceptability of the system with target elderly users and caregivers through structured testing and feedback collection.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, which is appropriate for the development and evaluation of innovative IT artefacts-in this case, a wearable fall-risk prediction system.

3.1 Research Approach

The research progresses through the following phases:

- Problem identification and literature review
- System design and hardware selection
- Data collection (sensor data from wearable device)
- Machine learning model development and training
- XAI integration and explanation generation
- System integration and prototype development
- Evaluation and validation

3.2 Technical Methods and Algorithms

- Sensor fusion: Accelerometer (3-axis) + Gyroscope (3-axis) data combined for rich movement representation
- Feature extraction: Signal windowing, statistical features (mean, variance, skewness, RMS)
- Classification model: Long Short-Term Memory (LSTM) network for sequential pattern recognition
- Explainability: SHAP (SHapley Additive exPlanations) for feature attribution
- Ensemble validation: Random Forest as a comparative baseline

3.3 Tools and Platforms with Justification

Tool / Platform	Purpose	Justification
Arduino / Raspberry Pi	IoT edge device	Low power, widely used, community support
MPU-6050 Sensor	Accelerometer +Gyroscope	Cost-effective,accurate, compact
Python (TensorFlow/Keras)	ML model development	Industry standard for deep learning
SHAP Library	Explainability	Best-in-class XAI,model-agnostic
Firebase / MQTT	Data transmission	Real-time IoT data protocols
Flask / React Native	Mobile alert interface	Cross-platform, lightweight

3.3.1 Frontend Design and User Experience (UX) Strategies

The user interface of the proposed fall-risk detection system is a critical component that bridges complex machine learning predictions with human comprehension. A well-designed front end ensures that caregivers and medical professionals can interpret real-time predictions instantly, act on alerts without ambiguity, and trust the system's outputs. This section documents the design philosophy, UI architecture, component layout, and UX principles derived from the system. prototype

3.4 Data Collection and Ethical Considerations

The integrity, validity and ethical soundness of the data collected for training and validating the fall-risk prediction model are foundational to the success of this research. This section details the data collection strategy, the datasets employed, preprocessing procedures and the ethical framework governing all data activities involving human participants.

3.5 Validation Strategy and Performance Metrics

Rigorous validation is essential to demonstrate that the proposed fall-risk prediction system performs reliably under real-world conditions and meets the clinical standards required for deployment in elder care settings. This section outlines the multi-layered validation strategy, the statistical performance metrics used to evaluate the model, the validation datasets and splits applied and the end-to-end system-level testing approach

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and its Components

The proposed system consists of four primary components operating in a pipeline:

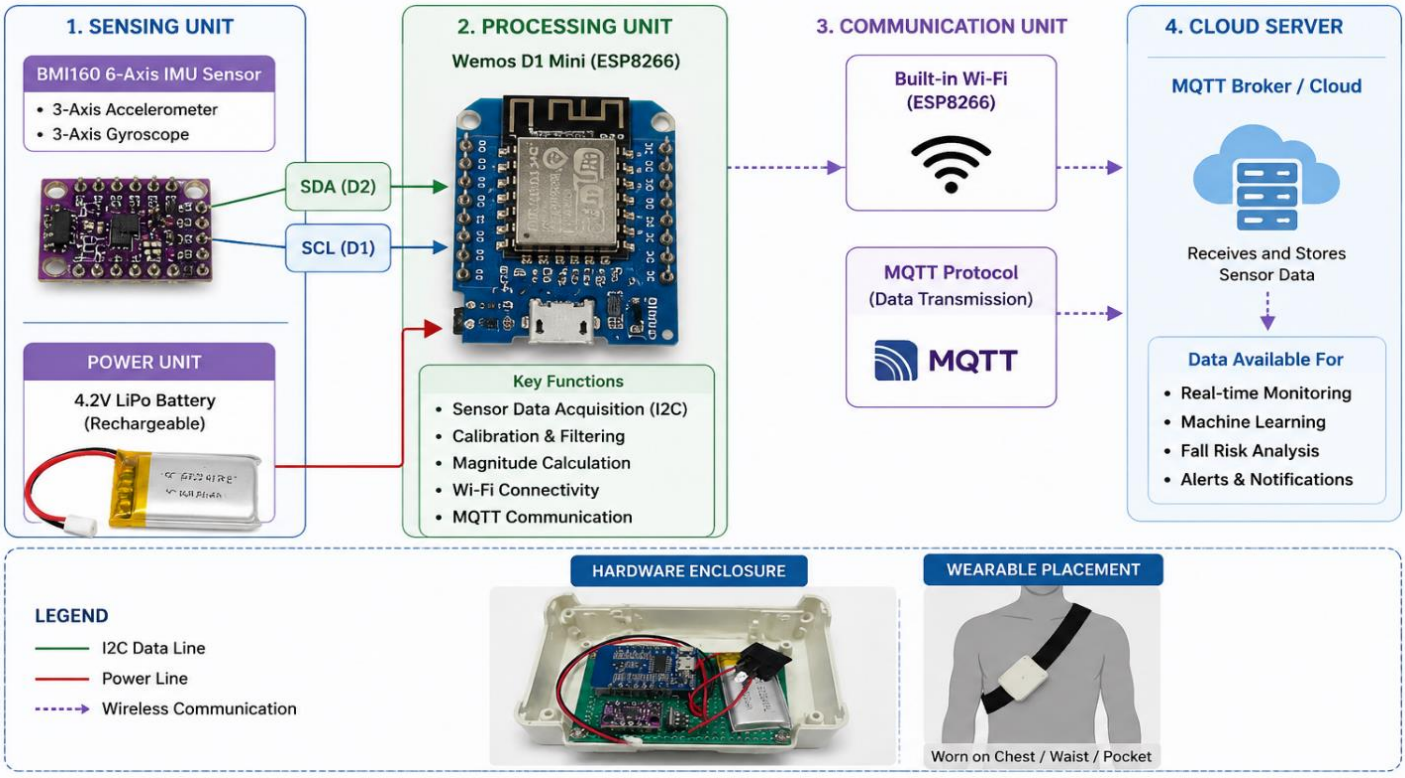
- **Wearable IoT Layer**: The MPU-6050 sensor (accelerometer + gyroscope) is mounted on a wearable device worn on the wrist or waist. It continuously captures 3-axis acceleration and angular velocity data at 50 Hz.

- **Edge Processing Layer:** A microcontroller (Raspberry Pi Zero / Arduino Nano 33 BLE) preprocesses the raw sensor data — applying signal windowing, noise filtering, and feature extraction — before transmitting to the prediction layer.
- **ML Prediction Layer:** The trained LSTM model receives feature vectors and outputs: (a) activity classification (walking/sitting/standing), (b) fall direction prediction (left/right/front/back), (c) confidence score (0–100%), and (d) stability check flag.
- **Explainability and Alert Layer:** The SHAP module generates a human-readable explanation of the prediction. If fall risk exceeds a threshold, an alert with explanation is sent to the caregiver mobile application via MQTT/Firebase.

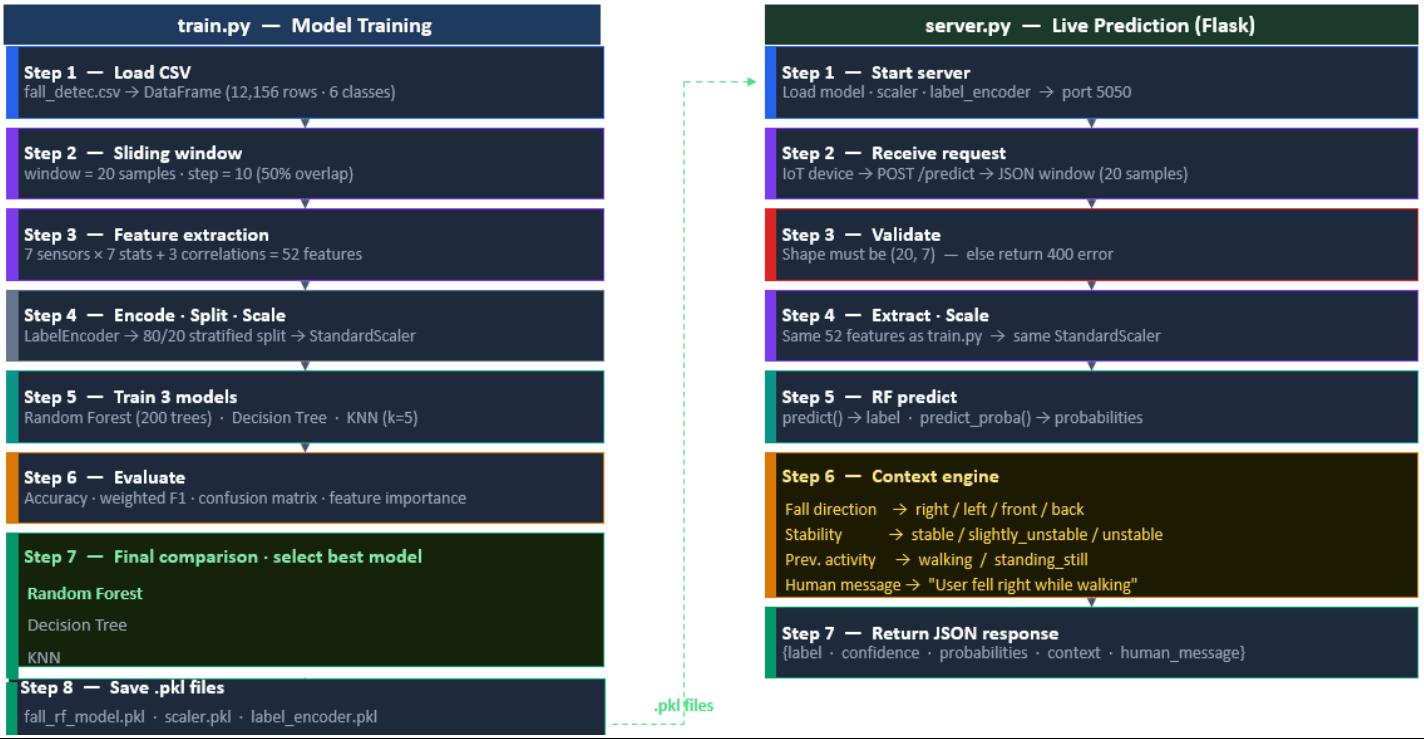
Scalability: The cloud layer supports multiple wearable devices, enabling deployment in care homes with multiple residents. Security is ensured through encrypted MQTT communication and user authentication in the mobile application.

4.2 System Diagram of Components

IoT Based Hardware System Diagram



End to End ML Pipeline



5. USER REQUIREMENTS

- Elderly users (primary wearers of the device)
- Family caregivers and relatives
- Medical professionals and physiotherapists
- Healthcare facility administrators

5.2 Functional Requirements (FR)

FR Number	Requirements
FR-01	The system shall continuously monitor the wearer's body movement using IoT sensors.
FR-02	The system shall classify the current activity state (walking, sitting, standing) in real time.
FR-03	The system shall predict the probability of an imminent fall before it occurs.
FR-04	The system shall predict the likely direction of the fall (left, right, front, back).
FR-05	The system shall provide a confidence score (0-100%) for each fall-risk prediction.
FR-06	The system shall monitor whether the individual stabilizes after an at-risk posture event.
FR-07	The system shall generate a human-understandable explanation for each prediction.
FR-08	The system shall send real-time alerts to caregivers via a mobile application.
FR-09	The system shall log historical movement and prediction data for clinical review.

5.3 Non-Functional Requirements (NFR)

NFR Number	Category	Requirement
NFR-01	Performance	Prediction latency shall not exceed 500 milliseconds from sensor reading.
NFR-02	Accuracy	Fall prediction accuracy shall achieve a minimum of 85% on test datasets.
NFR-03	Reliability	The device shall operate continuously for a minimum of 8 hours on a single charge.
NFR-04	Usability	The wearable device shall weigh less than 80g and be non-intrusive for daily use.
NFR-05	Security	All data transmission shall be encrypted using TLS 1.3.
NFR-06	Maintainability	System software shall be updatable remotely via OTA (Over-the-Air) updates.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary market: Elderly individuals (65+) living independently or in assisted living facilities.
Secondary market: Healthcare institutions, elder care homes and family caregivers of elderly individuals

6.2 Value Proposition

The system provides the world's first proactive, explainable wearable fall-risk predictor — preventing falls before they happen, reducing injury, and giving caregivers actionable, understandable insights to intervene early.

6.3 Revenue Model

- **Hardware sales**: Wearable device unit (one-time purchase)
- **SaaS subscription**: Monthly/annual subscription for cloud monitoring and analytics dashboard
- **Institutional licensing**: Bulk licensing for elder care homes and hospitals

6.4 Cost Estimation (Initial Phase)

Item	Quantity	Unit Cost(LKR)	Total(LKR)	Justification
MPU-6050 Sensor	3	500	1,500	Primary motion sensing; spares for development
Raspberry Pi Zero W	2	4,500	9,000	Edge computing and model inference
Arduino Nano 33 BLE	2	3,500	7,000	Lightweight MCU for sensor reading
LiPo Battery (3.7V 2000mAh)	3	1,200	3,600	Wearable power source

3D Printed Enclosure	2	1,500	3,000	Wearable form factor housing
Cloud Services (Firebase)	12 months	1,000	12,000	Real-time data storage and alerts
Miscellaneous (wires, PCB)	-	-	5,000	Prototyping components
		Total	41,100	

6.5 Pricing Strategy

Tier	Target	Pricing(Estimated)
Individual	Personal use / family caregivers	LKR 15,000 device + LKR 500/month subscription
Institutional	Care homes / hospitals	Custom enterprise licensing

6.6 Competitive Advantage

- Only solution combining proactive prediction + directional fall analysis + XAI explanation
- Lower cost than existing commercial systems targeting developing markets
- Interpretable outputs increase clinical adoption and trust

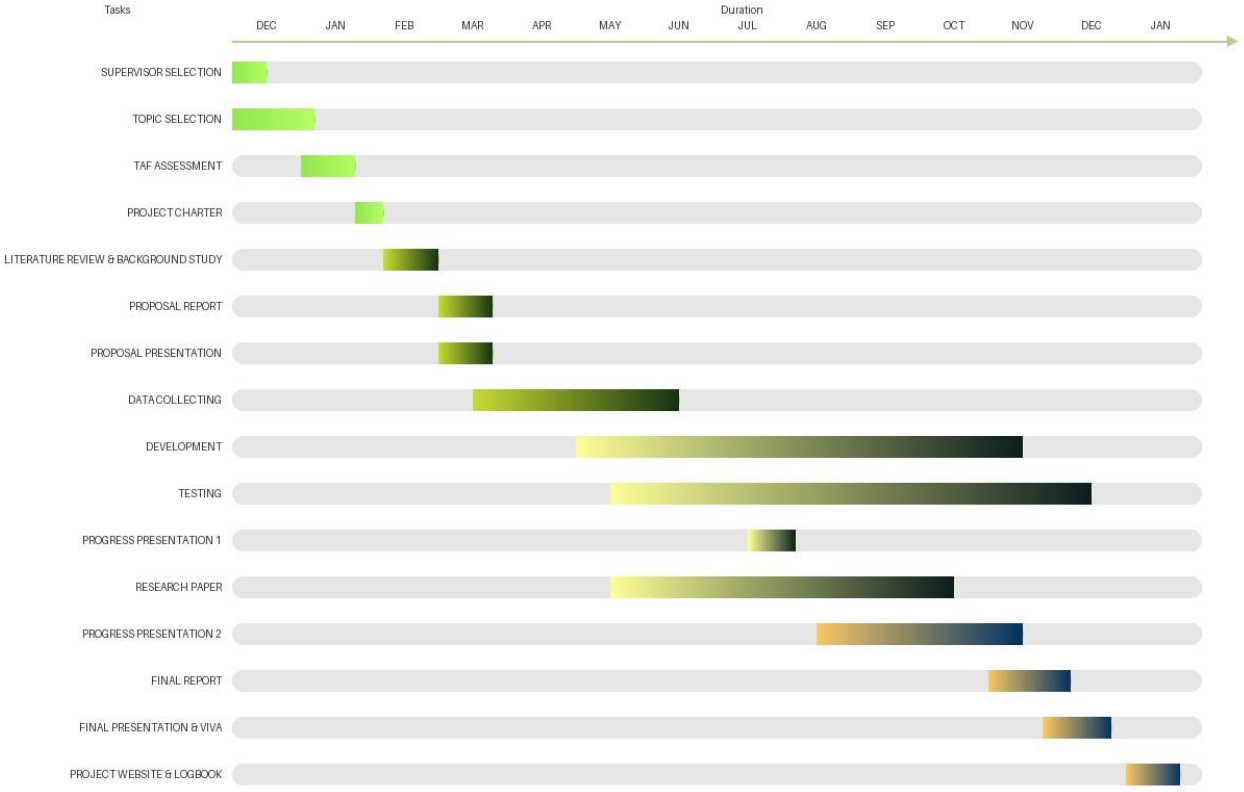
6.7 Intellectual Property (IP) Considerations

The novel combination of directional fall prediction and XAI explanation generation will be explored for patent protection. The ML model architecture and XAI integration pipeline are candidates for provisional patent filing.

7. WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task	Deliverable	Duration
1. Planning	Problem definition, literature review	Literature review report	Weeks 1-3
2. Design	System architecture, hardware selection	System design document	Weeks 4-5
3. Hardware Dev	Sensor integration, IoT device assembly	Functional wearable prototype	Weeks 6-8
4. Data Collection	Sensor data gathering, dataset preparation	Labelled movement dataset	Weeks 9-10
5. ML Development	Model training, validation, tuning	Trained ML model (>85% accuracy)	Weeks 11-14
6. XAI Integration	SHAP integration, explanation generation	Explainable prediction module	Weeks 15-16
7. Integration & Testing	Full system integration, end-to-end testing	Integrated system prototype	Weeks 17-18
8. Evaluation	User testing, performance benchmarking	Evaluation report	Weeks 19-20
9. Documentation	Final report, presentation preparation	Final dissertation	Weeks 21-24

8.GANTT CHART



9.REFERENCES

[1] X. Yang *et al.*, "Development and Validation of Machine Learning Models for Predicting Falls Among Hospitalized Older Adults: Retrospective Cross-Sectional Study," *JMIR Aging*, vol. 9, p. e80602, Jan. 2026. [Online]. Available: https://aging.jmir.org/2026/1/e80602

[2] A. Khunteta and A. Sharma, "Fall Detection for The Elderly People," *International Journal of Basic and Applied Sciences*, vol. 14, no. 3, pp. 45–58, Sep. 2025.

Exercise-Based Prevention & Clinical Screenings

[3] K. Narola and A. Bachkaniwala, "A REVIEW on Balance Enhancing Exercise Programme(Beep) on Balance and Quality of Life in Elderly Individuals," *Journal of Society of Indian Physiotherapists*, vol. 10, no. 2, pp. 88–94, Jul. 2026.

[4] J.-G. Ji and Y.-S. Kwak, "Effects of Fall Prevention Exercise on Fall-Related Physical Fitness and Quality of Life in Elderly Women," *The Asian Journal of Kinesiology*, vol. 27, no. 4, pp. 112–121, Oct. 2025.

[5] A. Supinganto, R. Mulyoto, N. K. Metri, and M. S. Hidayatullah, "Effectiveness of Multicomponent Elderly Exercise on Balance and Fall Risk: SKILAS-based Screening in West Lombok Elderly Schools," *Jurnal Ilmiah STIKES Yarsi Mataram*, vol. 16, no. 1, pp. 24–35, Jan. 2026.

[6] G. L. Rosa and R. M. Laranjeira, "The effects of Tai Chi Chuan on fall prevention in the elderly: Systematic Review: Tai Chi Chuan and fall prevention in the elderly," *Manual Therapy Posturology & Rehabilitation Journal*, vol. 24, pp. 1–12, Apr. 2026.

Wearable & Microcontroller Systems

[7] N. Ramakrishna, V. Saketha, K. H. Chary, and V. D. Kumar, "Smart Human Fall Detection System with Automated Safety Net and Emergency Response," *International Journal on Science and Technology*, vol. 12, no. 2, pp. 134–145, Apr. 2026.

[8] P. Burde, S. Bobade, P. Brahmankar, and D. Sharma, "AI-Powered Fall Detection Systems: A Solution for Elderly Safety and Well-Being," *Journal of Geriatric Care and Technology*, vol. 8, no. 4, pp. 201–213, Nov. 2025.

[9] M. S. Santos, J. P. Silva, and R. C. Mendes, "An ESP32 and MPU6050 Based Wearable System for Real-time Fall Detection and Alerting," *IEEE Sensors Journal*, vol. 24, no. 8, pp. 4512–4521, Apr. 2024.

[10] L. Wang, T. Zhang, and X. Liu, "A Wearable Tri-Axial Accelerometer System for Fall Risk Assessment in Community-Dwelling Older Adults," *IEEE Transactions on Biomedical Engineering*, vol. 70, no. 11, pp. 3120–3129, Nov. 2023.

Vision & Edge AI Systems

[11] S. N. Ramesh, P. D. Rathika, M. Sarkar, and C. Paolini, "Stereo Vision-Based Fall Prediction and Detection using Human Pose Estimation on the AMD Kria K26 SOM," *arXiv preprint arXiv:2606.07432*, Jun. 2026.

[12] M. Fayad, M. A. Merzoug, A. Mostefaoui, and R. Yahiaoui, "A lightweight AI system for real-time elderly fall detection," *Computing*, vol. 108, no. 4, pp. 789–807, Apr. 2026.

[13] Y. Zhou, X. Sun, and Z. Li, "Privacy-Preserving Vision-Based Fall Detection in Sensitive Indoor Environments Using Skeleton Evolution Features," *IEEE Transactions on Multimedia*, vol. 27, pp. 1423–1435, Feb. 2025.

[14] H. Kim and J. Park, "Edge-Computing Framework for Real-Time Fall Detection Using Quantized Deep Neural Networks," *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 8450–8461, May 2024.

Non-Invasive & Ambient Sensors

[15] P. Muruganandham, F. F. Sahapudeen, S. Jayaraman, and S. K. P. Nagaraju, "Deep learning-based localization and tracking of elderly individuals for healthcare monitoring," *Discover Artificial Intelligence*, vol. 6, no. 2, Art. no. 45, Jun. 2026.

[16] J. Smith and R. Jones, "Passive Infrared (PIR) Sensor Arrays for Localization-Aware Fall Detection in Bathroom Environments," *IEEE Journal of Biomedical and Health Informatics*, vol. 29, no. 1, pp. 112–123, Jan. 2025.

[17] R. Caldwell and M. Amin, "Radar-Based Fall Detection Using Joint Time-Frequency Analysis and Deep Residual Networks," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 3, pp. 1890–1902, Jun. 2024.

Machine Learning & Predictive Analysis

[18] F. Martinez, A. Gomez, and L. Silva, "A Comparative Study of Machine Learning Algorithms for Fall Prediction Models on the cStick Dataset," *Journal of Medical Systems*, vol. 49, no. 2, pp. 74–86, Oct. 2025.

[19] T. Nguyen and K. Wong, "Predicting Fall Likelihood in Geriatric Patients Using Electronic Health Records and Behavioral History Manipulation," *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 12, pp. 301–311, Mar. 2024.

[20] S. Patel, H. Rahman, and A. Al-Majeed, "Data Normalization Strategies (Min-Max vs Z-Score) in Enhancing Lightweight Neural Network Architectures for Fall Detection," *Pattern Recognition Letters*, vol. 178, pp. 45–52, Jan. 2024.

10.APPENDICES

Appendix A – Risk Analysis

Risk	Likelihood	Impact	Mitigation
Sensor data noise affecting accuracy	Medium	High	Apply Kalman filtering and data augmentation
Insufficient training data	Medium	High	Use public benchmark datasets + data augmentation
Battery life shorter than expected	Low	Medium	Optimize firmware; select higher capacity battery
User discomfort with wearable	Medium	Medium	Conduct early usability testing and iterate design
Ethical clearance delay	Low	High	Apply for clearance in first research week

Appendix B – Ethical Considerations

- All human participant data will be collected with written informed consent.
- Participant data will be anonymized and not linked to personal identifiers.
- Participants may withdraw from the study at any time without consequence.
- Data will be stored securely and deleted after research completion.
- Institutional ethics committee approval will be obtained prior to any data collection.